ECE 6276 - Spring 2025
Lab 1: Barrel Shifter

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Last Modified: February 3rd 2025

Simulation Waveform

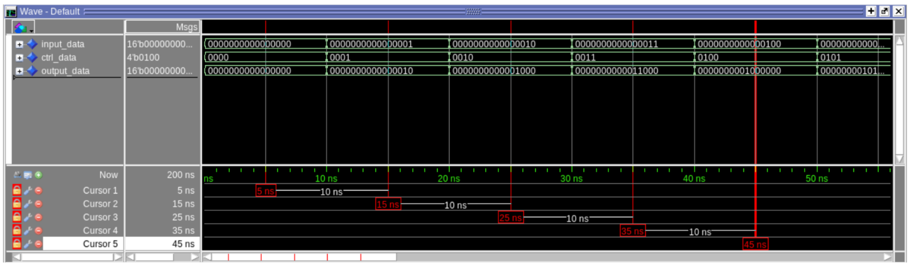

Figure 1: The waveforms of test cases 1 to 5

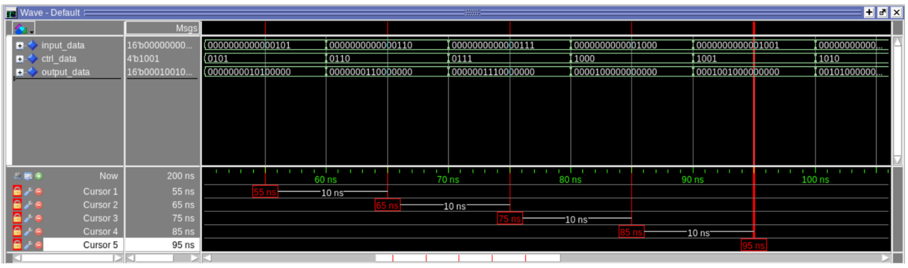

Figure 2: The waveforms of test cases 6 to 10

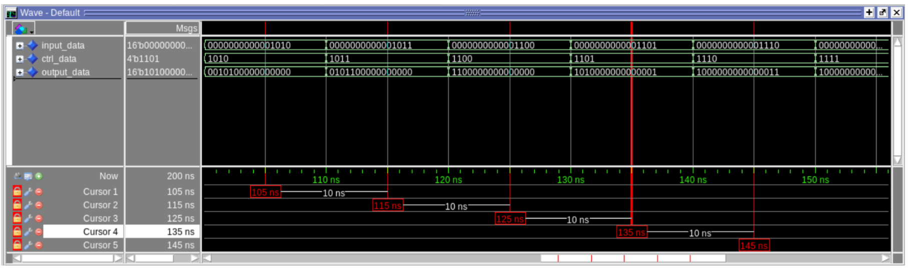

Figure 3: The waveforms of test cases 11 to 15

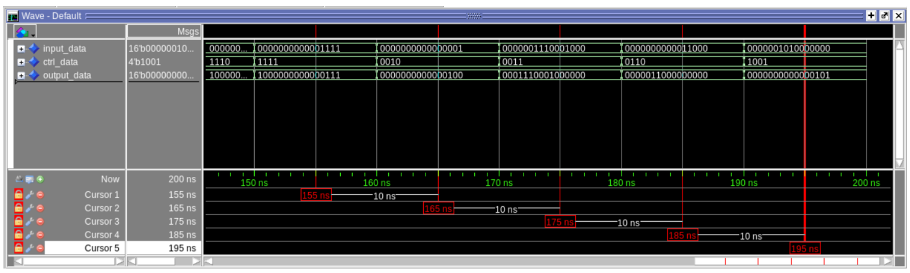

Figure 4: The waveforms of test cases 16 to 20

test case	input_data	ctrl_data	output_data
1	0000000000000000	0000	0000000000000000
2	0000000000000001	0001	0000000000000010
3	0000000000000010	0010	0000000000001000
4	0000000000000011	0011	0000000000011000
5	0000000000000100	0100	0000000001000000
6	0000000000000101	0101	0000000010100000
7	0000000000000110	0110	0000000110000000
8	0000000000000111	0111	0000001110000000
9	0000000000001000	1000	0000100000000000
10	0000000000001001	1001	0001001000000000
11	0000000000001010	1010	0010100000000000
12	0000000000001011	1011	0101100000000000
13	0000000000001100	1100	1100000000000000
14	0000000000001101	1101	1010000000000001
15	0000000000001110	1110	1000000000000011
16	0000000000001111	1111	1000000000000111
17	0000000000000001	0010	0000000000000100
18	0000001110001000	0011	0001110001000000
19	0000000000011000	0110	0000011000000000
20	0000001010000000	1001	0000000000000101

Table 1: Simulation Results

Manual Verification

1. When the input data is 0000 0011 1000 1000, and ctrl data is 0011, the output data should be rotated left by 3 bits, i.e. 0001 1100 0100 0000, that is the same as the output: **0b0001 1100 0100 0000**.
2. When the input data is 0000 0000 0001 1000, and ctrl data is 0110, the output data should be rotated left by 6 bits, i.e. 0000 0110 0000 0000, that is the same as the output: **0b0000 0110 0000 0000**.
3. When the input data is 0000 0010 1000 0000, and ctrl data is 1001, the output data should be rotated left by 9 bits, or rotated right by 7 bits, i.e. 0000 0000 0000 0101, that is the same as the output: **0b0000 0000 0000 0101**.

Implemetation Summary

Resource	Utilization	Available	Utilization %
LUT	32	20800	0.15
IO	36	106	33.96

Table 2: Resource Utilization

On-Chip	Power (W)	Power (%)	Used
Slice Logic (LUT)	0.166	2	32
Signals	0.370	4	52
I/O	8.597	94	36
Static Power	0.128	-	-
Total	9.262	-	-

Table 3: Power Consumption

Verification Strategies

1. Function Verification
   - Simulate the barrel_shifter.vhd design using ModelSim to verify correctness.
   - Compare simulated waveforms with expected results from manual calculations.
   - Use testbenches to apply various input test cases and observe corresponding outputs.
2. Test Case Coverage
   - Verify both the minimal shift (0 bit shift) and maximum shift (full width bit shift)
   - Use gtID-based test cases to validate correct behavior with specific values.
3. Debug and Log Anysis
   - Monitor signals in waveform view (input_data, ctrl_data, output_data) to ensure correct transformations.
   - Utilize cursors and timing markers in ModelSim to track delays and potential issues.
   - Report to compare the expected and actual results on the terminal.
   - Record the output data, and analyze the result with manual verification
4. Synthesis Validation
   - Run synthesis in Xilinx Vivado to analyze both Resource utilization and Power Consumption